

1. Stealth Scan (SYN Scan)

- **Command:** `nmap -sS <target>`
- **Description:** Sends a **SYN** packet (like initiating a TCP handshake), but doesn't complete the handshake.
- **How It Works:**
 - Open port → Target replies with **SYN-ACK**, Nmap sends **RST**.
 - Closed port → Target replies with **RST**.
- **Purpose:** Fast and stealthy; avoids full connections and often bypasses basic logging.

◆ 2. Xmas Scan

- **Command:** `nmap -sX <target>`
- **Description:** Sends a packet with **FIN, URG, and PSH** TCP flags set — resembles a "lit up" packet.
- **How It Works:**
 - Open port → No response (per RFC).
 - Closed port → RST.
- **Limitation:** Only works on **Unix-like systems**, Windows systems usually ignore it.

◆ 3. Null Scan

- **Command:** `nmap -sN <target>`
- **Description:** Sends TCP packets with **no flags set**.
- **How It Works:**
 - Open port → No response.
 - Closed port → RST.
- **Limitation:** Windows does not follow RFC rules for these scans — it always sends RST.

◆ 4. FIN Scan

- **Command:** `nmap -sF <target>`
- **Description:** Sends a TCP packet with only the **FIN** flag.
- **How It Works:**
 - Open port → No response.
 - Closed port → RST.
- **Purpose:** Helps **bypass firewalls and IDS** that only expect SYN scans.

◆ 5. Timing and Performance

- **Command:** `nmap -T<0-5> <target>`

- **Purpose:** Controls scan speed vs. stealth trade-off.

- **Timing Levels:**

Level Name	Use Case
T0	Paranoid IDS evasion (very slow)
T1	Sneaky Stealthy, slow
T2	Polite Reduces bandwidth usage
T3	Normal Default
T4	Aggressive Faster scanning, riskier
T5	Insane Maximum speed, least stealth

- **Example:** `nmap -sS -T4 <target>`

◆ 6. MAC Spoofing Scan

- **Command:** `nmap --spooof-mac <value> <target>`
- **Description:** Changes the **MAC address** of the scanner interface.
- **Types of Spoofing:**
 - Specific MAC: `--spooof-mac 00:11:22:33:44:55`
 - Vendor: `--spooof-mac Apple` (uses a MAC prefix from Apple)
 - Random: `--spooof-mac 0` or `--spooof-mac random`
- **Example:** `nmap -sS --spooof-mac Cisco <target>`
- **Purpose:** Hides real identity or mimics another vendor/device.

◆ 7. Decoy Scan

- **Command:** `nmap -D decoy1,decoy2,...,<yourIP> <target>`
- **Description:** Uses **fake IP addresses** to confuse IDS and logs.
- **Example:** `nmap -D 10.0.0.1,10.0.0.2,ME <target>`
- **Alternative:** Random decoys: `-D RND:5`
"ME" tells Nmap which IP is yours.

◆ 8. MTU Change in Scan

- **Command:** `nmap --mtu <value> <target>`
- **Description:** Sets a **custom MTU (Maximum Transmission Unit)** for packets.
- **Use Case:** Can help **evade detection** or work around specific network issues.
- **Example:** `nmap --mtu 16 <target>`
- **Note:** Use MTU values in **multiples of 8**.

◆ 9. Fragmentation in Nmap

- **Command:** `nmap -f <target>`
- **Description:** **Fragments** IP packets into small pieces to evade detection by firewalls or IDS.
- Sends the scan data in small IP packets (e.g., 8 bytes) to **obfuscate the scan signature**.
- **Example:** `nmap -sS -f <target>`
- **Purpose:** Avoids deep packet inspection and some intrusion detection systems.

◆ 10. Zombie Scan (Idle Scan) using msfconsole

- **Nmap Command (standalone):**

```
nmap -sl <zombie_IP> <target>
```

- **In msfconsole:**

1. Start Metasploit:

```
msfconsole
```

2. Use idle scan module:

```
use auxiliary/scanner/portscan/idle
```

3. Set options:

```
set RHOSTS <target>
```

```
set ZOMBIE <zombie_IP>
```

```
run
```

- **Description:**
 - Uses a "**zombie**" host (idle system with predictable IP ID values) to scan the target.
 - The scan appears to come from the **zombie**, hiding the attacker's real IP.
- **Requirement:** Zombie must not generate its own traffic, and must have **predictable IP ID sequences**.

Scan open ports for known vulnerabilities.

- **Use Nmap NSE scripts:**

```
nmap --script=vuln -p <port> <ip>
```

✅ Summary Table

Scan Type	Command Example	Evasion Capability	Works On
SYN (Stealth)	<code>nmap -sS <target></code>	Medium stealth	All systems

FIN	<code>nmap -sF <target></code>	Firewall evasion	Unix/Linux
Null	<code>nmap -sN <target></code>	Firewall evasion	Unix/Linux
Xmas	<code>nmap -sX <target></code>	Firewall evasion	Unix/Linux
Decoy	<code>nmap -D RND:5 <target></code>	Source identity hiding	All systems
MAC Spoofing	<code>nmap --spoof-mac Apple <target></code>	Identity hiding	All systems
Fragmentation	<code>nmap -f <target></code>	IDS/IPS evasion	All systems
MTU Change	<code>nmap --mtu 16 <target></code>	IDS/IPS evasion	All systems
Idle (Zombie) Scan	<code>nmap -sl zombieIP <target></code>	Ultimate stealth	Requires idle host